Title	**CAAP: Bridging the Gap Between Academia and Accessibility**
Course	CS/QTM/LING-329: Computational Linguistics
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Abstract

This paper presents Capture Assistant in Academic Papers (CAAP), a comprehensive tool designed to aid both students and researchers in academia by facilitating the understanding of complex academic papers. CAAP integrates keyphrase extraction techniques with large language models (LLMs) to provide contextual explanations for key concepts within academic papers. By leveraging a term frequency comparison algorithm and KeyBERT for keyphrase extraction, and OpenAI's GPT-3.5 for definition generation, CAAP offers users an accessible interface to view key concepts and understand technical terminology.

The main merit of CAAP lies in its innovative approach to integrating information from the figures in academic papers with textual analysis, thereby enriching the keyphrase extraction process. Additionally, CAAP's ability to seamlessly execute keyphrase extraction and definition generation in a single workflow enhances efficiency and usability, representing a significant advancement in academic research tools.

CAAP has the potential to democratize access to academic knowledge by making research literature more accessible and comprehensible to a wider audience. By empowering students and researchers to navigate complex technical jargon and identify key concepts within academic papers, CAAP aims to foster a more inclusive and collaborative research community, ultimately advancing the frontiers of knowledge and innovation.

1 Introduction

Every year, academia produces a plethora of research papers spanning various topics. While these papers are primarily intended for researchers within their respective domains, they also serve as valuable resources for college students interested in academia and research. However, unlike seasoned researchers with expertise in their fields, many college students often struggle with deciphering the technical jargon and complex concepts embedded within academic papers. In addition, researchers who seek to submit their papers to specific conferences and archives may struggle to find relevant keywords that may wholly define the content of their papers.

To address such concerns, we propose the Capture Assistant in Academic Papers (CAAP). CAAP is a tool designed to aid users in extracting key phrases in academic papers while providing a web-based platform for key phrase explanations that is accessible and comprehensible to all levels of expertise. Our tool is intended for the following user groups:

1. Undergraduate and graduate students who seek to understand the contents of a research paper more easily
2. Researchers in academia who desire an automatic method to extract and define keywords
3. Databases that wish to facilitate search-based algorithms with keywords for locating similar studies

Our tool comprises two integral components: a keyphrase extraction feature and a user interface that can capture and display the extracted keyphrases directly from the papers. The keyphrase extraction aspect of CAAP utilizes a term frequency document to extract the highlighted key concepts. In tandem, KeyBERT is utilized to bolster the number of keyphrases that are extracted from the document.

In addition to the keyphrase extraction tool, the CAAP tool incorporates a component dedicated to defining and addressing the difficult keyphrases that users may encounter while reading the papers. By leveraging the text summarization ability of large language models (LLMs), the component defines each keyphrase in accordance with predefined level of difficulty that the users can adjust. Drawing from the content of the given academic paper, our tool is equipped to provide contextually relevant explanations to clarify these keyphrases regarding ideas or terminology within the papers.

CAAP enables students and researchers to grasp the main ideas of the papers and provide definitions of the terms they aim to understand without delving into extensive research or grappling with specialized vocabulary. The merits of this project are as follows:

1. We propose an unsupervised method of keyphrase extraction, utilizing a term frequency-based algorithm to highlight the main technical concepts of academic papers.

2. We utilize the visual aspects of academic papers, including figures, to further enhance the process of keyphrase extraction.

3. We present an accessible interface that conducts both keyphrase extraction and term summarization to assist users in understanding technical concepts in academia.

2 Background

The primary function of CAAP utilizes an area of natural language processing (NLP) called keyphrase extraction (KE). KE, first explored in the 1990s, is the process in which important phrases or words are highlighted automatically in a given document. [11] During the first explorations of KE, Turney [11] suggested learning and non-learning algorithms, including decision trees and the developed GenEx algorithm, for handling this particular task.

Today, the methods for implementing KE are largely divided into two categories. The first is through supervised learning, where models are utilized to facilitate the extraction process. Song et al. [8] and Papagiannopoulou and Tsoumakas [7] assert that with the development of neural network and deep learning models, much of KE has been advanced using such models. These models are identified as classification models, and as they are trained on annotated data, demonstrate comparable results [12].

Pre-trained language models, in particular, have helped facilitate the process even further. Due to their state-of-the-art performances in various NLP categories, pre-trained language models are strong candidates for optimizing keyphrase extraction. [13, 5, 8] As a result, KE models also utilize pre-trained language models such as BERT and RoBERTa as their base. [9]

In addition to supervised learning methods, various unsupervised methods have also been introduced to facilitate KE. From an unsupervised approach perspective, KE can be considered a graph problem. [4, 3] Thus, several word graph-based methods have been proposed, some of which outperform other traditional methods such as TF-IDF and Bayesian Network-based methods, but do not seem to produce comparable results compared to pre-trained language models. [6]

One of the more recent studies conducted by Bennani-Smires et al. [1] proposed an unsupervised approach of KE using sentence embeddings. This approach demonstrated a diverse set of keyphrases that users were more satisfied with in comparison to other unsupervised approaches. Even more recently, a more statistical method has been proposed, one in which the statistics of a singular corpus are utilized to extract the keywords of a document. [2]. Results have demonstrated that this method outperforms most state-of-the-art unsupervised methods for KE and may show comparable results against supervised methods as well. [2]

Compared to such unsupervised methods, our approach leverages term frequency analysis to conduct keyphrase extraction. Utilizing a term frequency document that Extracted keywords from various computer science documents, our approach focuses on the terms within the provided document to highlight the key concepts within it. In addition, previous works in keyphrase extraction involve the text of a document, but utilizing the images of a document to contribute to the document's contents has been underexplored. Thus,

by integrating the visual information of documents into textual analysis, our methods offer a novel approach to using figures of papers in tandem with text for keyphrase extraction.

3 Methodology

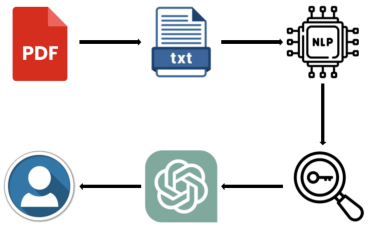


Figure 1: Workflow diagram.

Figure 1 demonstrates a basic workflow of the components of CAAP. Since many academic papers are archived as PDFs, we develop a PDF information extracting system that extracts the information of the given document as raw text.

One of the main components of the PDF extraction system is its multimodal function which can identify and process information in figures such as images and tables. The system extracts any data within the figures and converts them into text. The output of the processed document is a string of the document's raw text, inclusive of that in figures. The system employs the Tesseract OCR (Optical Character Recognition), which, by leveraging machine learning models, extracts characters from images. Preprocessing is executed on the text to ensure the removal of stop words and unrelated vocabulary, such as numeric data.

3.1 Keyphrase Extraction

Our term frequency-based algorithm and KeyBERT are used for keyphrase extraction. The term frequency algorithm utilizes a term frequency dictionary. If the dictionary is missing, it can be generated using a collection of papers from any academic domain, thus making the dictionary specialize in the particular area of academia.

The term frequency document is generated using a sample number of academic papers, specifically focusing on the computer science domain. For this research, approximately 200 computer science-based documents are utilized to create this term frequency document. Once generated, the algorithm compares each individual word's frequency in the new paper to the same word's frequency from the dictionary of the 200 computer science papers to take the top k terms with the highest ratio. This provides a set of keywords that represent the most frequent terms within the input document relative to other computer science papers. For our research, we choose to extract the top 5 terms using this algorithm.

Furthermore, the keyphrase extraction utilizes KeyBERT to extract additional top words from the document. These two results are concatenated, where thereafter duplicate keywords are automatically merged, and in cases where one keyword is a substring of another, the shorter one is retained while the longer keyword is removed. The output is a set of up to 10 unique keywords extracted by the two methods and processed from the academic paper.

3.2 OpenAI GPT pipeline

LLMs are used for providing explanations of the keyphrases to different user groups. Three levels of difficulty — novice, proficient, and expert — are identified. After obtaining the top keywords extracted from the document and identifying the level of difficulty in explaining the keywords, GPT-3.5 is used to define each keyword with respect to the context of the paper.

The prompt used for defining these keyphrases is as follows:

```
You are a great dictionary fluent in technical jargon from academic papers.  Use
this paper to help define the terms:
{the text of the paper}
At (a novice, beginner/an intermediate, proficient/an expert, intricate) level please
define {keyword}
```

Each keyword and its associated definition are paired together in a dictionary, and both are returned on the website.

3.3 Flask

Since CAAP is built on Python, we utilize Flask, a popular web development framework adapted for Python, to deliver a lightweight, flexible, and extensible front-end web app. We design a "Home" page, an "About Us" page, and a functional page for users to upload their papers. The "Home" page and the "About Us" page consist of a user guide and an information page about the team.

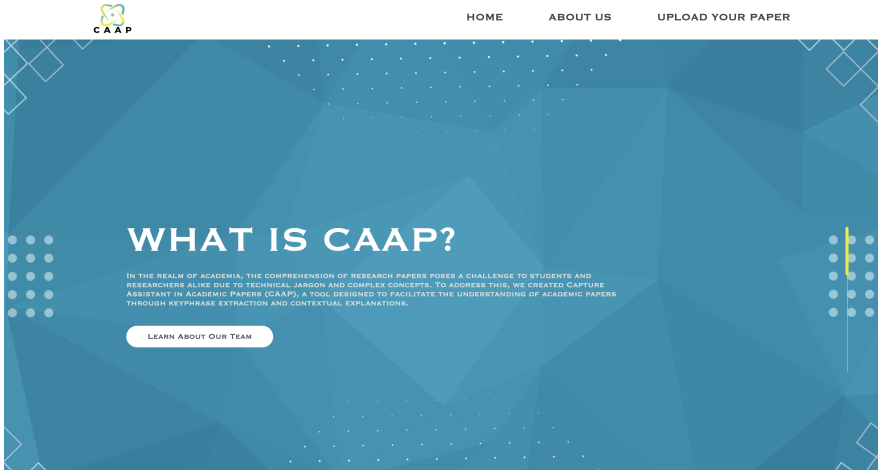


Figure 2: A screenshot of CAAP's home page.

The "Upload Your Paper" page allows users to upload their academic papers in PDF format. The file will then be sent to the first two steps in the pipeline, where PDF processing and keyphrase extraction take place. Lastly, the users can view the results or the explanations of the keywords directly on the web app after the GPT responses are sent back.

Figure 2 demonstrates a screenshot of the local webpage, with Figure 3 exemplifying the tool being used to extract keyphrases from Toxtli et al. [10]'s paper on the development of a task-oriented chatbot.

4 Evaluation

CAAP is evaluated on its two features: keyphrase extraction and definition generation. To achieve this, we collected an evaluation set of 16 academic papers in the domain of Computer Science and other research published in Economics, Political Science, and Sociology.

The evaluation set is composed of 10 Computer Science proceeding papers from various ACM conferences: CHI, ICMLC, ASPLOS, and BCB. We also use papers in Economics, Political Science, and Sociology from Arxiv, the American Journal of Political Science, and Sage journals to evaluate CAAP's functionalities in other areas of academia. These papers cover a range of topics from Deep Learning Algorithms, LLM Inference, and Sustainability to Voting, Climate Action, and Community Interviews.

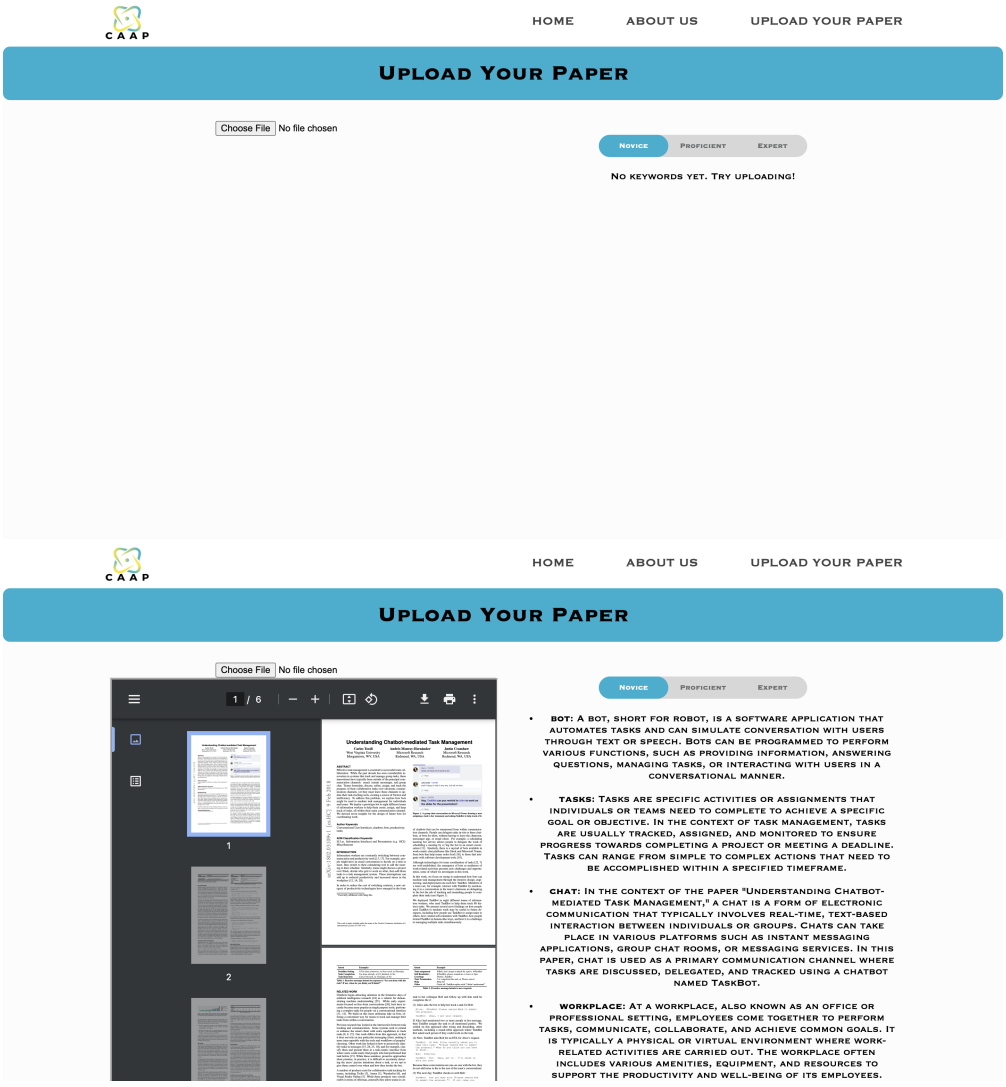


Figure 3: Screenshot of CAAP's keyphrase extraction tool at use. An academic paper is uploaded to the tool, in which the webpage displays the uploaded paper and the associated keywords along with their definitions, defined at the desired level of difficulty.

4.1 KE Evaluation

The selected papers in Computer Science have keywords predefined by the authors for references. To evaluate CAAP's capabilities of KE on these papers, we use the predefined keywords as the golden standard and calculate the accuracy of CAAP's keyphrase extraction using the F_1 score.

The keywords for papers in Economics, Political Science, and Sociology are not identified. As a result, we utilize GPT-4 to identify keywords for those papers without predefined keywords. Its output is used as the golden standard to calculate the F_1 score for CAAP's results.

The following prompt for GPT-4 is used to identify the keywords for the papers without predefined keywords:

```
Can you identify the keywords in this paper?
```

The F_1 score for CAAP's KE accuracy is calculated as follows:

$$F_1 = \frac{TP}{TP + 0.5(FP + FN)} \tag{1}$$

where **TP** is the number of true positives, or number of CAAP's predicted keywords that align correctly with the golden standard (predefined keywords by the paper's authors or GPT-4); **FP** is the number of false positives, number of predicted keywords that do not align with golden standard; **FN** is the number of false negatives, or keywords from the golden standard that were not captured by CAAP.

The **TP**, **FP**, and **FN** are manually identified as there may be captured keywords that have similar semantic meaning yet differing lexical orthography.

4.2 Definition Generation Evaluation

To evaluate the definition generation of CAAP, we followed a structured approach utilizing similarity scores. Firstly, for each paper in the evaluation set, we assess the Novice and Expert levels of definitions generated by the GPT pipeline. Subsequently, we calculate the cosine similarity score between the Novice and Expert levels to quantify their similarity. It is assumed that the Novice level definitions would generally exhibit simpler language and content compared to the Expert level, hence resulting in a lower similarity score between them.

5 Results and Discussion

Table 4 demonstrates the accuracy of KE by CAAP on each academic paper in the evaluation set. The highest F_1 score reaches 0.6, showcasing notable performance. While the average F_1 score is approximately 0.36, indicating room for improvement, it's noteworthy that CAAP successfully captures a significant number of keywords in each document, especially considering the utilization of a term frequency document rather than a supervised method.

Table 5 displays the similarity scores between the Novice and Expert level definitions for the extracted keywords. The average Cosine Similarity Score between the Novice and Expert level definition is approximately 0.78, which establishes higher similarity between the two definitions despite the established difference in level of difficulty.

The potential issue leading to these similar definitions might be due to factors related to GPT and the prompts used to generate these definitions. The prompts used for the GPT pipeline may not be detailed enough for GPT to generate highly differential definitions for Novice and Expert levels. For improvement, we may continue prompt-engineering to elicit more diverse results.

5.1 Limitations

Our approach faces several limitations, notably in terms of complexity and adaptability. While KeyBERT is a powerful model that effectively extracts keyphrases, it poses computational challenges, particularly on less

Paper Category	F1 Scores for the keyword capture's accuracy
CS CHI	0.25
CS CHI	0.29
CS CHI	**0.6**
CS ICMLC	0.2
CS ICMLC	0.36
CS ICMLC	0.29
CS ASPLOS	0.29
CS ASPLOS	0.5
CS ASPLOS	0.42
CS BCB	0.31
Economics	0.33
Economics	0.31
Political Science	0.18
Political Science	0.57
Sociology	0.36
Sociology	0.55

Figure 4: F_1 Scores for CAAP's Accuracy in KE

Paper Category	Cosine Similarity Score
CS CHI	0.88
CS CHI	0.49
CS CHI	0.67
CS ICMLC	0.75
CS ICMLC	0.93
CS ICMLC	0.87
CS ASPLOS	0.92
CS ASPLOS	0.83
CS ASPLOS	0.83
CS BCB	0.91
Economics	0.85
Economics	0.91
Political Science	0.11
Political Science	0.75
Sociology	0.86
Sociology	0.85

Figure 5: Cosine Similarity Scores between Novice and Expert level Definitions for Keywords.

powerful machines. When executing the extraction on a laptop, documents of moderate length — around six to eight pages – can take about 15-20 seconds to fully process, while longer documents exceeding ten pages may require over a minute for full processing.

Another limitation lies in the quality and adaptability of the term frequency document. In specialized fields like computer science, where new concepts emerge frequently, the term frequency document may require frequent regeneration to reflect these updates accurately. Additionally, our current approach does not adequately capture keyphrases from diverse academic areas, leading to overly general keyphrase extraction, particularly when applied to papers from different academic domains.

Furthermore, while our method achieves notable accuracy in keyword extraction, leveraging a term frequency document rather than supervised methods may not always result in the most precise keyword identification. The predefined keywords, especially by the authors, often rely on summarizing the key ideas of the document, rather than extracting the top-k keyphrases directly.

6 Conclusion

Through this project, we introduce CAAP, a comprehensive tool designed to facilitate the understanding of academic papers by extracting keyphrases and providing contextual explanations. Our approach combines keyphrase extraction using a term frequency document and KeyBERT with text summarization capabilities of LLMs to generate user-adjustable definitions for extracted keyphrases. Through evaluation, we demonstrate CAAP's effectiveness in accurately extracting keyphrases and providing definitions across various domains of academia.

CAAP demonstrates potential to significantly impact both students and researchers in academia. By simplifying the process of deciphering complex technical jargon and identifying key concepts within academic papers, CAAP empowers users to engage more effectively with research literature, thereby fostering greater comprehension and knowledge dissemination.

Future directions to direct this research may include the enhancement of scalability and adaptability of our keyphrase extraction approach. More efficient algorithms that offer resolutions to computational complexities may be considered. Additionally, there may be methods to update the term frequency document while executing keyphrase extraction to accommodate evolving concepts and keyphrases in the academic domain. Furthermore, optimization of the definition generation component, such as fine-tuning the prompts used for LLMs, can enhance the clarity and relevance of the provided explanations.

CAAP represents a promising step towards democratizing access to academic knowledge and fostering a more inclusive and accessible research community. By addressing the challenges associated with understanding complex academic papers, CAAP has the potential to revolutionize the way individuals engage with research literature, ultimately advancing the frontiers of knowledge and innovation.

Our work is publicly available on Github.

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